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Noncritical birefringence phase-matched second harmonic generation in a lithium-niobate-on-insulator micro-waveguide for green light emission

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Lithium niobate on insulator (LNOI) holds great potential for frequency conversion, where a variety of high-performance nonlinear devices based on different structures has been demonstrated. Here, we report on second harmonic generation (SHG) in MgO-doped LNOI ridge micro-waveguides for efficient green light emission, via an exact type-I noncritical birefringence phase matching (BPM). The LNOI micro-waveguide has a cross section of $\sim 3 \times 4 \mu\text{m}^2$, featuring low coupling loss with lens fiber. The normalized conversion efficiency from a continuous-wave (cw) pump to its second harmonic is measured to be $37\%/W\text{cm}^2$ in a single-pass configuration. The device shows both relatively high efficiency and a void of periodic poling, offering a potential solution for efficient and scalable green light sources and frequency converters. © 2024 Optica Publishing Group

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Lithium niobate (LN) is widely used in optics, especially in nonlinear photonics, because of its wide transparent window, large electro-optic and nonlinear coefficients, and other exceptional properties. The excellent crystal has long been established as a prevailing photonic material in optics and photonics, whose applications include optical telecommunication, optical sensing/imaging, atomic and molecular optics, and quantum information technology. However, dense integration in bulk LN is inaccessible for decades due to lack of waveguide structures with high refractive index contrast. The development of a lithium niobate on insulator (LNOI) or a thin-film lithium niobate (TFLN) has greatly promoted the renaissance of LN-based integrated photonics [1–10]. The strong light confinement in LNOI waveguides remarkably enhances the light-matter interaction strength and has brought photonic devices with unprecedentedly high performance. Efficient frequency converters based on

second-order nonlinearity, e.g., SHG, sum-/different-frequency generation, and optical parametric oscillation/amplification, have pivotal importance across a wide variety of science and applications. These nonlinear wave mixing processes have been demonstrated on the platform with impressively high efficiency in both LNOI and TFLN waveguides. Many research groups have reported SHG in LNOI and TFLN through different phase matching schemes, such as birefringence phase matching (BPM) [11,12], modal phase matching (MPM) [13–15], quasi-phase matching (QPM) [16–19], cyclic QPM [20,21] and others [22,23], in ridge waveguides and microresonators. Among them, the poling-free MPM method is an easy-to-implement yet effective solution. Previously reported demonstrations are usually achieved at the cost of small mode overlapping between fundamental and higher-order spatial modes. The QPM method requires periodic poling (domain inversion) in the crystal which is technically demanding, and the cyclic QPM has only been realized in TFLN microresonators.

There has been a strong quest for compact, efficient, and fiber-compatible SHG devices for nonlinear photonics and applications, and recently, much effort has been dedicated to frequency conversion in the telecommunication bands. On the other hand, visible or short-wavelength lasers are highly desired in atomic, molecular, and optical physics. For example, efficient narrow-linewidth green light generation is highly desired for many applications. Biochemical sensing, holography, underwater communication, and quantum technology also benefit from laser sources at green wavelengths. However, the power from direct laser emission at short wavelengths is somehow limited. Frequency conversion is an appealing and feasible approach to solve this problem.

In this Letter, we report on efficient BPM SHG for green light emission (532 nm) in the MgO:LNOI ridge micro-waveguide void of periodic poling. The noncritical type-I (oo→e) BPM

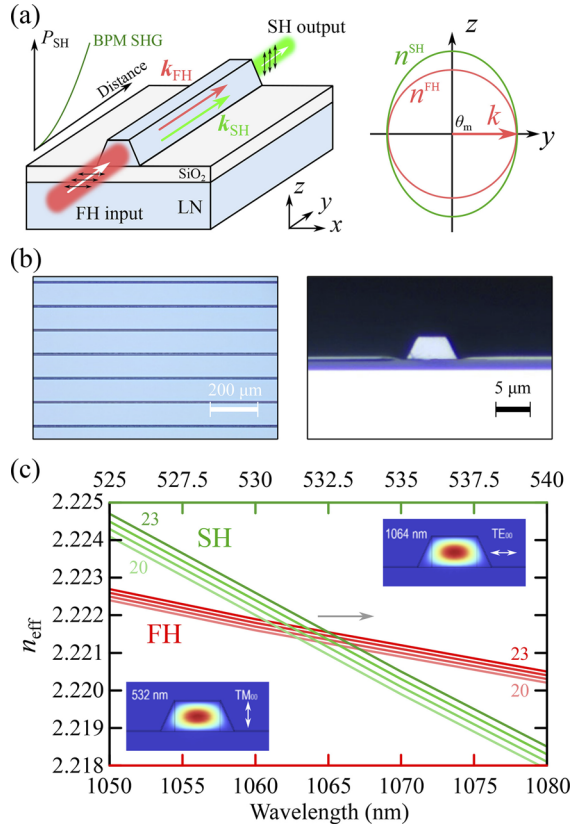


Fig. 1. (a) Illustration of noncritical BPM SHG in a LNOI waveguide. (b) Microscopic images of the LNOI micro-waveguide array and the waveguide end facet. (c) Evolution of the effective indices of the FH (TE₀₀) and SH (TM₀₀) guided modes at different temperatures (20, 21, 22, and 23°C), showing BPM wavelength at around 1064 nm in the waveguide. The BPM wavelength redshifts as the temperature increases. Inset: simulated electric field distribution of the FH and SH modes. The arrow indicates the polarization direction.

condition is satisfied between fundamental transverse modes in the waveguide with large mode overlapping. We experimentally achieve a normalized conversion efficiency of 37%/Wcm² for cw laser (1064 nm) pumping in a single-pass frequency-doubling configuration. The noncritical BPM is suitable for long waveguides, giving rise to high overall performance even though d_{31} of LN is utilized. The SHG into the green wavelength is found to be stable, where the photorefractive effect is well mitigated by MgO doping.

The noncritical BPM for frequency conversion of fundamental harmonic (FH) to its second harmonic (SH) in the LNOI ridge micro-waveguide is schematically depicted in Fig. 1(a). In the z -cut, y -propagating waveguide, the effective refractive indices of FH (fundamental TE mode) and SH (fundamental TM mode) are designed to be equal. The mode overlap factor is maximized. The FH and SH propagate collinearly at the noncritical BPM angle of $\theta_m = 90^\circ$. The spatial walk-off effect is not present under this condition. The condition is found to be met on the MgO-doped micrometer-thick LNOI platform, considering both the material and waveguide structure dispersion at the FH wavelength of 1064 nm. This is very close to the BPM short-wavelength limit of LN. The condition is not possible in TFLN nano-waveguides due to strong structural dispersion, where MPM is usually applied at the

cost of sacrificed mode overlapping. Highly efficient MPM SHG can still be achieved using novel strategies, like semi-nonlinear and dual-layered TFLN waveguides [24–26].

In our experiment, the frequency converter is fabricated from a commercial LNOI wafer (NANOLN) with a 3- μ m-thick z -cut MgO-doped LN layer bonded on top of a 2- μ m-thick silica buffering layer on a 400- μ m thick LN substrate. The LNOI micro-waveguide is fabricated using ultraviolet (UV) lithography and deep inductively coupled plasma (ICP) etching. The method, overcoming the high etch resistance of LN by using micrometer-thick UV mask, is able to deeply etch LN to form ridge micro-waveguides with smooth sidewalls. The advantage of this method is that it is cost-effective and suitable for scalable production. The obtained micro-waveguide is compatible with optical fiber, possessing intrinsically low coupling loss with lens or high-NA fiber. The fabrication procedure is similar to the one described in Ref. [27]. Figure 1(b) shows the optical microscopic image of the fabricated micro-waveguide array sample, as well as the cross section of the ridge micro-waveguide for the SHG process. In this work, we further increase the etching depth. The top LNOI layer is deeply etched to an averaged etching depth of 2.8 μ m, and the etching sidewall angle is 65°. The over-etching (etch trench) has helped to form a free-standing MgO:LNOI ridge waveguide on SiO₂. The ridge micro-waveguide has a top width of 3.6 μ m and a length of 2 cm. The facets of the waveguide are diced and optical-grade polished but without anti-reflection coating. The sample was fabricated at the Center for Advanced Electronic Materials and Devices, SJTU. Figure 1(c) presents the simulated effective refractive indices of the TE₀₀ mode at 1064 nm (i.e., FH) and the TM₀₀ mode at 532 nm (i.e., SH) according to the micro-waveguide dimensions. The two modes share similar mode distribution, contributing to a large mode overlapping. The calculated mode overlap factor is 0.99, much larger than that in the TFLN nano-waveguides. The dispersion is calculated at different temperatures of 20–23°C. The crossing of the lines indicates the noncritical BPM condition at 1064 nm at around the room temperature.

Firstly, the linear optical properties of the fabricated MgO:LNOI micro-waveguide is characterized. The propagation loss of the micro-waveguide is evaluated at the telecommunication band, due to the lack of continuously tunable laser sources at 1064 nm in our experiment. The propagation loss measurement setup is shown in Fig. 2(a). A cw telecommunication tunable laser (1520–1600 nm), whose power is controlled by a variable optical attenuator, is used as the source. The polarization is controlled by a polarization controller. The laser light is then fed into the LNOI micro-waveguide via a lens fiber, which has a nominal beam size of 2.5 μ m at the focus. The output light is then collected by another lens fiber and monitored using a photodiode or a powermeter. The experimentally measured insertion loss is 3.5 dB for TE and 4.0 dB for TM input at 1550 nm. The propagation loss is evaluated using the Fabry–Perot (F–P) interference method. Each end facet of the waveguide acts as a partial reflecting mirror, leading to an FP-like cavity in the waveguide. Propagation loss α can be extracted from the transmission (or reflection) spectrum, i.e., FP interference fringes as in Fig. 2(b) by scanning the wavelength. The free spectral range (FSR) is measured to be 26.5 pm for TE and 27.5 pm for TM light, which is expected to be 26.4 pm and 27.4 pm, respectively. The value of the propagation loss is calculated to be 0.3 dB/cm and 0.41 dB/cm for TE and TM modes, respectively. The propagation loss is on par with that using optical-grade dicing methods [28]. The

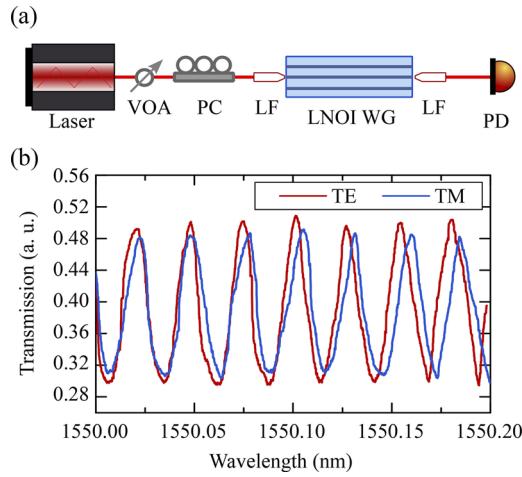


Fig. 2. (a) Schematic of the propagation loss measurement setup. VOA, variable optical attenuator; PC, polarization controller; LF, lens fiber; WG, waveguide; PD, photodiode. (b) Measured transmission FP interference pattern for the TE and TM polarized input.

coupling loss is then estimated to be approximately 1.5 dB/facet. Low coupling loss is attained even without anti-reflection coating because of good mode matching between the lens fiber and the micro-waveguide. The insertion loss of the micro-waveguide at 1064 nm is measured to be 3.2 dB for TE and 3.5 dB for TM input, respectively. Thus, considering similar coupling loss, the propagation loss at 1064 nm is believed to be similar to that measured at 1550 nm.

The main part of the experimental setup for the single-pass frequency-doubling in the MgO:LNOI micro-waveguide is shown in Fig. 3(a). The cw 1064-nm laser (linewidth: <0.01 nm) is launched into the fiber and coupled into the micro-waveguide via a lens fiber. The light polarization is controlled by an in-line polarization controller. The sample is mounted on a three-dimensional nanostage for precise positioning and a thermoelectric cooler (TEC) for temperature control. The waveguide output is collimated in free space using an optical lens and dispersed by a prism. The FH and SH output is well spatially separated and filtered for monitoring, using a powermeter and a spectrometer. The noncritical BPM SHG is realized in our micro-waveguide at the room temperature (22°C), well consistent with the theoretical prediction. Figure 3(a) also shows the experimentally observed bright green emission under moderate pumping (at 20-mW pump power). The FH and SH spectrum is shown in Fig. 3(b). The peak width is only limited by the resolution of the spectrometer (0.3 nm). The fundamental mode profiles of the FH and SH waves are confirmed by observing the output light spots in the far field, as shown in the inset of Fig. 3(b). The FH is viewed using an infrared sensor card (red). The round spot shape suggests the waves are guided at their fundamental transverse modes in the waveguide. Figure 3(c) presents the measured SH (green) power as a function of the FH input, showing good quadratic relationship at the small-signal condition. The experimental normalized conversion efficiency, defined as $\eta = P_{SH}/P_{FH}^2 L^2$, where P_{FH} and P_{SH} are the respective powers at the waveguide output and L is the waveguide length, is measured to be 37%/Wcm² in the small-signal approximation condition ($P_{FH} \leq 50$ mW in the experiment). For the 2-cm-long sample, the fiber-to-space conversion efficiency is about

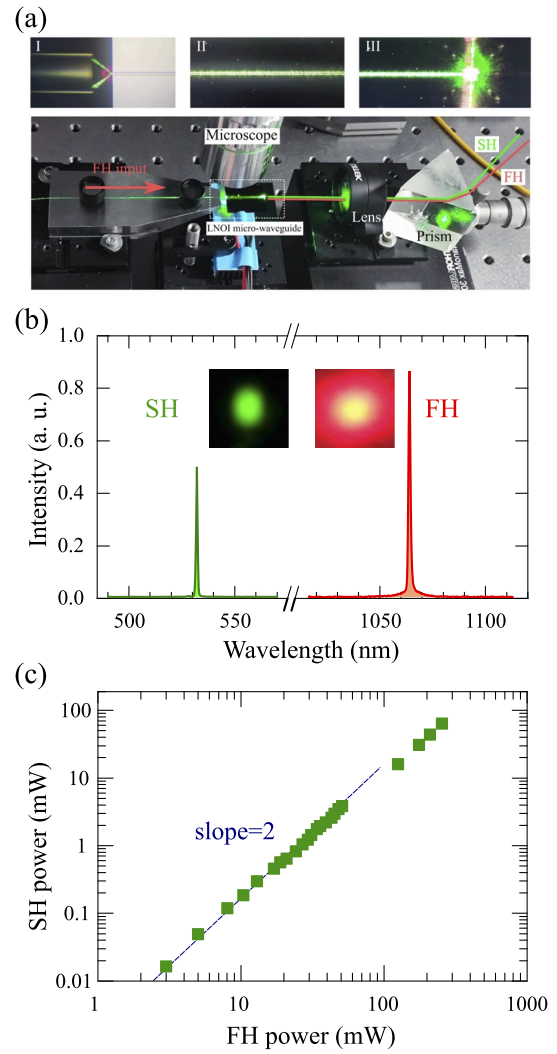


Fig. 3. (a) Experimental setup for SHG in the MgO:LNOI micro-waveguide. (I) Light coupling using a lens fiber. (II) SHG in the MgO:LNOI micro-waveguide. (III) Output SHG at the micro-waveguide end. (b) Spectrum of the FH and SH waves. (Not to scale.) Inset: FH and HS output spots. (c) Quadratic relationship between the FH and SH power, which starts to deviate at high input power as the pump is partially depleted.

150%/W for small inputs. Considering the coupling loss, the on-chip conversion efficiency is estimated to be 53%/Wcm², and the theoretical normalized conversion efficiency is 72%/Wcm² in the undepleted pump approximation. For higher input powers with pump depletion, an absolute conversion efficiency of 12.6% is obtained with 500-mW input. This is good enough for an intense cw laser or pulsed laser. Due to the power limit of our laser, SHG at higher pump intensities has not been investigated. However, there are a handful literature showing that (MgO- or ZnO-doped) LNOI ridge waveguides are capable of handling intense cw lasers up to several Watts. The performance of our device is on par with some previously reported LNOI ridge waveguides or TFLN nano-waveguides, whose length or coupling is limited [17,24,29,30]. Our BPM micro-waveguide well balanced the fiber coupling and nonlinear conversion efficiency, making the overall device performance also as good as its counterparts.

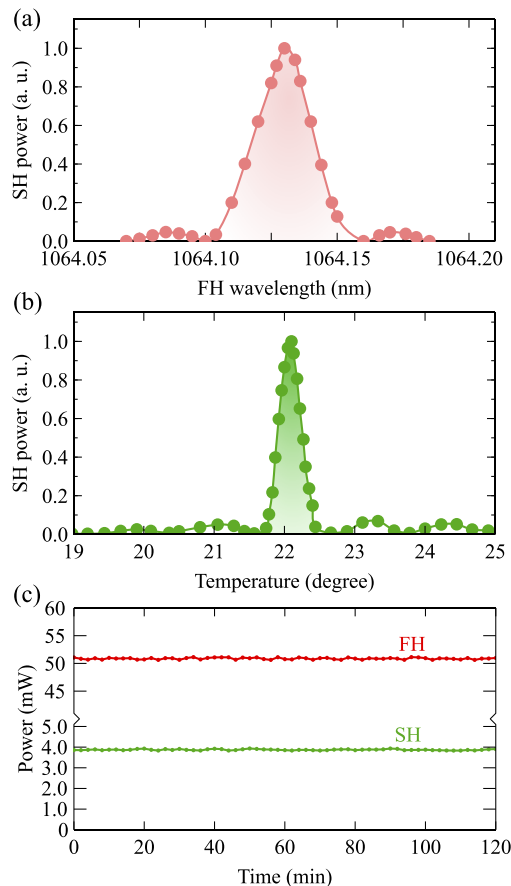


Fig. 4. (a) Normalized SH intensity as a function of the FH wavelength. (b) Normalized SH intensity versus temperature. (c) Stability test of SHG in the MgO:LNOI micro-waveguide.

Furthermore, the tolerance of the noncritical BPM is investigated by scanning the wavelength and temperature, as shown in Figs. 4(a) and 4(b), in consistent with the theoretical prediction (sinc^2 function). The 1064-nm laser in our experiment can be finely tuned (but discretely) in a narrow range of 0.7 nm. The full width at half maximum (FWHM) of the wavelength tolerance is 0.025 nm, which agrees with the theoretical prediction. The FWHM of the temperature tolerance is 0.5°C, which can be well stabilized using the temperature controller. Finally, we investigate the long-term stability of the frequency converter against the photorefractive effect. The photorefractive effect has to be considered when the LN device works in the short visible wavelengths, which would severely deform the wavefront and deteriorate the frequency-doubling process. The effect can be well mitigated via MgO doping. We monitor the SHG stability over a time period of 2 h. The result is shown in Fig. 4(c). The 2% variation of SH dominantly arises from the pump fluctuation (1% instability). This proves that our MgO:LNOI micro-waveguide can well convert NIR light

into visible range in a stable manner and is suitable for realistic applications.

In conclusion, we have demonstrated a noncritical type-I BPM in the MgO:LNOI micro-waveguide for efficient green light emission, which exhibits a conversion efficiency of 37%/Wcm² or 150%/W for the 2-cm-long waveguide. The micro-waveguide also shows good stability against the photorefractive effect. The scalable and fiber-compatible feature would make the LNOI micro-waveguide appealing for realistic applications.

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Data availability. Data underlying the results presented in this paper are not publicly available at this time but may be obtained from the authors upon reasonable request.

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